Why Models Hallucinate, and Can They Know When They Are Unsure?

Graduate AI-Internals Module Q11 (DLS 2026)

Deep Learning Systems

October 11, 2026

The Student's Question

"Why do models hallucinate, and can they know when they are unsure?"

- *Hallucination*: fluent, confident output that is factually wrong or unsupported.
- Two sub-questions, often conflated:
  1. Why does the *generation* produce falsehoods?
  2. Does the *internal state* encode a usable signal of uncertainty?
- Thesis: hallucination is partly **statistical and unavoidable**, partly **incentivized by training and evaluation**, and models **do carry** partial self-knowledge that we can extract.

Source 1: Objective Mismatch and the Density Trap

A language model minimizes cross-entropy over a corpus; it learns $p(\text{token} \mid \text{context})$, not truth.

- **Exposure bias / autoregression**: during generation the model conditions on its own (possibly wrong) outputs, compounding errors.
- **Long-tail facts**: rare entities have little support, so the model interpolates plausible-but-false completions.
- **(author?)** [1] show hallucination arises even for an ideal learner: generating calibrated-but-novel text necessarily incurs some rate of fabrication, linking it to density estimation limits.
- Lesson: a model optimized for likelihood will trade some correctness for fluency.

Source 2: Training and Evaluation Incentives

Hallucination is not only a bug—it is partly *rewarded*.

- **(author?)** [2] argue models hallucinate because standard benchmarks score like exams: guessing beats abstaining, so "I don't know" is penalized.
- RLHF can amplify confident assertion: **(author?)** [3] optimize for human-preferred answers, and humans often prefer decisive, fluent responses.
- **(author?)** [4] document *sycophancy*: models adjust answers toward the user's stated view, trading accuracy for approval.
- **Implication**: fix the scoring rule (reward calibrated abstention) and some hallucination disappears.

Source 3: Do Models Encode Their Own Uncertainty?

Evidence that the internal state *knows* more than the output admits.

- **(author?)** [5] find models can be trained/prompted to predict P(True) of their own answers and are often well-calibrated on whether they know.
- **(author?)** [6] show a probe on hidden activations detects whether a statement the model generates is true or false—internal "lie" signals exist.
- **(author?)** [7] discover latent truth directions *without labels* (CCS), suggesting an unsupervised uncertainty substrate.
- So: self-knowledge is partially present but not faithfully surfaced by greedy decoding.

Source 4: Calibration and How to Read It

Distinguish *knowing* from *saying you know*.

- **(author?)** [8] show modern neural nets are often *miscalibrated*—confidence $\neq$ accuracy; temperature scaling helps.
- **(author?)** [9] extend calibration study to pretrained transformers (in/out of domain).
- Verbalized confidence ("I'm 90% sure") and token log-probs are both imperfect proxies; RLHF can degrade log-prob calibration.

Operational definition
A model is calibrated if, among claims it reports at confidence c, a fraction $\approx c$ are correct: $\mathbb{E}[\mathbf{1}(\text{correct}) \mid \hat{p} = c] = c$.

Source 5: Detecting Hallucination at Inference

Practical signals that approximate "am I unsure?"

- **Sampling consistency**: **(author?)** [10] (SelfCheckGPT) flag hallucinations by measuring disagreement across sampled generations—no external DB needed.
- **Semantic entropy**: **(author?)** [11] cluster answers by meaning and measure entropy over clusters, a better uncertainty estimate than token entropy.
- **Retrieval grounding**: **(author?)** [12] (RAG) reduce fabrication by conditioning on retrieved evidence, shrinking the long-tail problem.

Rule of thumb: high semantic entropy or low cross-sample agreement $\Rightarrow$ treat the claim as unsupported.

Synthesis: A Causal Picture

Why hallucinate?

- Likelihood $\neq$ truth [1]
- Error compounding in autoregression
- Scoring rewards guessing [2]
- RLHF/sycophancy [4]

Can they know?

- Partial self-knowledge [5]
- Internal truth signals [6, 7]
- But miscalibrated output [8]
- Recoverable via entropy/consistency [10, 11]

Hallucination $\approx$ statistical inevitability + misaligned incentives; uncertainty exists internally but is lost at the output.

Takeaways and Exam Pointers

1. Hallucination has *multiple* causes; never attribute it to a single bug.
2. Models carry a partial, probeable signal of their own reliability—extracting it is the open engineering problem.
3. Mitigations stack: retrieval grounding, semantic-entropy gating, calibrated abstention, and better evaluation scoring.
4. **Be precise**: "confident" (output) $\neq$ "certain" (internal) $\neq$ "correct" (world).

Next lab: implement semantic-entropy detection and compare against verbalized-confidence calibration.

References

[1] A. T. Kalai and S. S. Vempala. Calibrated Language Models Must Hallucinate. *Proc. ACM Symposium on Theory of Computing (STOC)*, 2024.

[2] A. T. Kalai, O. Nachum, S. S. Vempala, and E. Zhang. Why Language Models Hallucinate. *arXiv:2509.04664*, 2025.

[3] L. Ouyang et al. Training Language Models to Follow Instructions with Human Feedback. *NeurIPS*, 2022.

[4] M. Sharma et al. Towards Understanding Sycophancy in Language Models. *arXiv:2310.13548*, 2023.

[5] S. Kadavath et al. Language Models (Mostly) Know What They Know. *arXiv:2207.05221*, 2022.

[6] A. Azaria and T. Mitchell. The Internal State of an LLM Knows When It's Lying. *Findings of EMNLP*, 2023.

References ii

[7] C. Burns, H. Ye, D. Klein, and J. Steinhardt. Discovering Latent Knowledge in Language Models Without Supervision. *ICLR*, 2023.

[8] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On Calibration of Modern Neural Networks. *ICML*, 2017.

[9] S. Desai and G. Durrett. Calibration of Pre-trained Transformers. *EMNLP*, 2020.

[10] P. Manakul, A. Liusie, and M. J. F. Gales. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. *EMNLP*, 2023.

[11] L. Kuhn, Y. Gal, and S. Farquhar. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. *ICLR*, 2023.

[12] P. Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*, 2020.